Bridge Day 6 INSTRUCTOR KEY

Multi-Step Calculations, Summaries, and Conditional Output

Learning sequence: Predict - Trace - Explain - Test - Interpret

Monday, July 13, 2026 • 20 points • target 17 minutes

Name		**UIN**		**Section**	

Assessment boundary and evidence source

Contract: 07a04826c975
Cumulative foundations: input conversion, arithmetic, min/max/sum/len, Boolean decisions, f-strings
Scaffold level: complete geometry and classification program
Evidence source: the matching v5 workbook readiness/evidence pages and VIP activities 18.8, 18.9, 18.10, 18.11, 18.12.
Show intermediate state for trace credit. Program credit requires one coherent solution, a stated assumption when the prompt leaves one implicit, and at least one test whose expected result can distinguish correct from incorrect logic.

Item 1. Evaluate ceiling-style kit arithmetic. - 4 points

```
groups = 7
items_per_group = 5
spares = 4
items = groups * items_per_group + spares
kits = (items + 11) // 12
leftover_space = kits * 12 - items
print(kits, leftover_space)
```

Question: Show the item total, adjusted numerator, quotient, and exact output.

Model response: items is 7 * 5 + 4 = 39. Adding 11 before floor division gives 50 // 12 = 4 kits. Four kits hold 48 items, leaving 9 unused spaces. Exact output: 4 9.

Item 2. Trace a numeric summary and formatting boundary. - 4 points

```
values = [4.5, 7.0, 2.5, 6.0]
total = sum(values)
average = total / len(values)
print(min(values), max(values), f"{average:.2f}")
```

Question: Compute every summary and give exact output, including the formatted average.

Model response: The total is 20.0 across four values, so average is 5.0. The minimum is 2.5 and maximum is 7.0. Two-decimal formatting produces exact output 2.5 7.0 5.00.

Item 3. Combine arithmetic with a conditional label. - 4 points

```
mass = 18.0
purity = 84.5
cost = 12 + 1.8 * mass
hold = purity < 85 or mass > 25

if hold:
    label = "hold"
else:
    label = "accepted"

print(f"{label}:  {cost:.2f}")
```

Question: Show the cost calculation and each Boolean component before giving exact output.

Model response: The cost is 12 + 1.8 * 18 = 44.4. purity < 85 is True and mass > 25 is False, so the or result hold is True. Exact output: hold: 44.40.

Complete program - 8 points

- Read length, width, and height as floats.
- Print invalid if any dimension is not positive.
- Otherwise compute volume and rectangular-prism surface area.
- Classify oversize when volume is above 500 or the largest dimension is above 12; otherwise classify standard.
- Print Volume and Surface area to two decimals, followed by Class: and the label.

Program workspace

Model program

```
length = float(input())
width = float(input())
height = float(input())

if length <= 0 or width <= 0 or height <= 0:
    print("invalid")
else:
    volume = length * width * height
    surface_area = 2 * (length * width + length * height + width * height)
    if volume > 500 or max(length, width, height) > 12:
        label = "oversize"
    else:
        label = "standard"
    print(f"Volume:  {volume:.2f}")
    print(f"Surface area:  {surface_area:.2f}")
    print(f"Class:  {label}")
```

Test input, expected result, and why this test matters

Reasoning and test evidence The validity branch protects all formulas. Surface area keeps the three face-pair terms visible before doubling their sum. For 10, 5, and 4, volume is 200 and surface area is 220, so the label is standard. For 13, 5, and 8, volume is 520 and the largest dimension is 13, so both independent checks support oversize.
Scoring guidance: 1 input, 1 validity, 2 volume/surface formulas, 2 compound classification, 1 exact precision/output, 1 boundary test.

Quiz evidence is sampled from the taught construct boundary; workbook and notebook evidence remain the primary learning surfaces.